RESEARCH · LLM EVALUATION · ORQ.AI

Weak judges, strong panel

An ensemble approach to LLM eval.

Arian Pasquali · orq.ai research · May 2026

TAKEAWAYS

Three things, if nothing else.

01

For the right reasons

Name the signal first: hard-case escalation, bias coverage, or ensemble accuracy. Can't name one? Ship a single judge.

02

Diversity is the product

Three small models, three providers — cheaper than one large judge, with uncorrelated errors. Correlated judges = 3× invoice.

03

Disagreement is the signal

Splits flag hard cases. Route them to humans. Judges that never disagree are one judge billed three times.

THE PROBLEM

One judge, several biases.

A single LLM judge isn't neutral. Different training data, architectures, instruction tuning, constitutional rules — these manifest as *systematic* bias.

DOCUMENTED

Self-preference

Judges over-rate outputs from their own model family. Yang et al. 2026.

DOCUMENTED

Position

In pairwise A-vs-B, order matters more than the answer. Shi et al. 2025.

DOCUMENTED

Length / verbosity

Longer answers win regardless of correctness. MT-Bench 2023.

DOCUMENTED

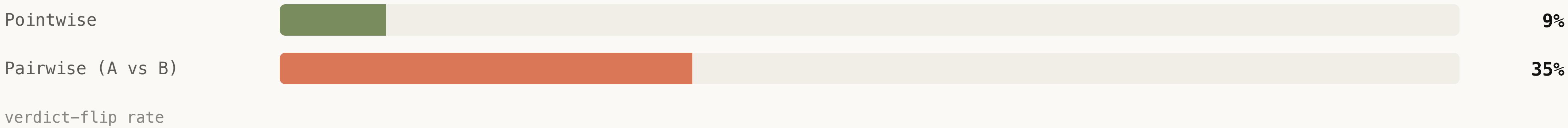
Confidence mismatch

High-confidence-trained models penalise nuanced "I'm not sure" replies.

DATA

Pairwise judges flip 4× more often than pointwise.

Same judge, same items, slightly perturbed prompts. The bias-reduction case for panels is strongest in pairwise mode.

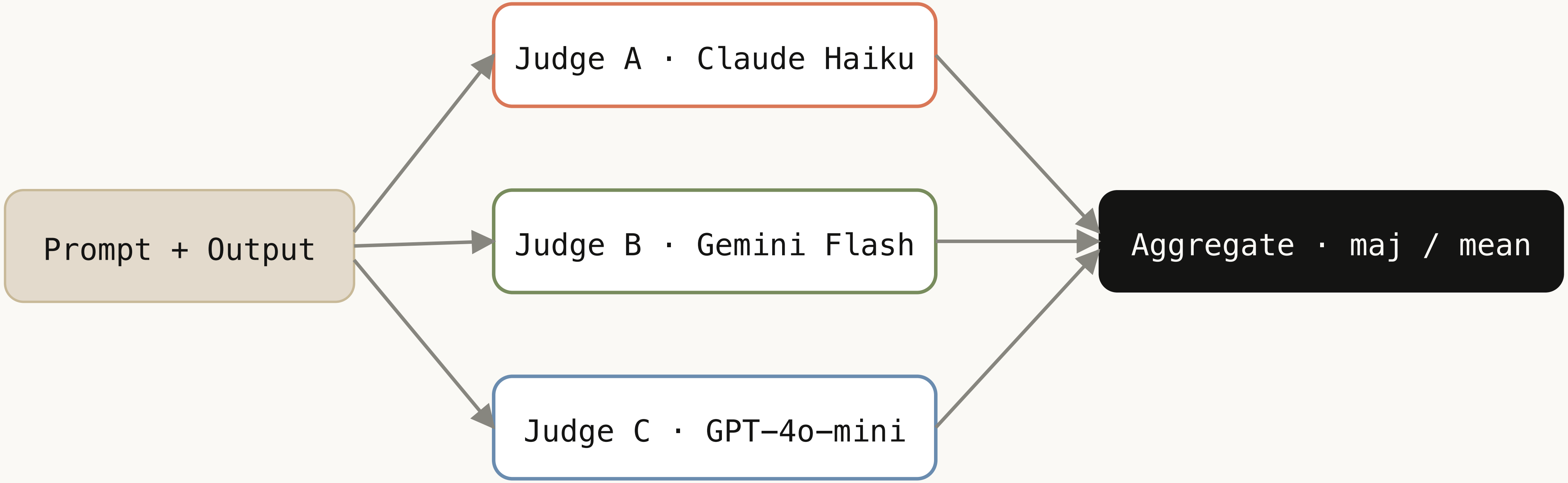


Wang et al. 2025. Pairwise prefs flip ~35% vs ~9% pointwise under benign re-prompts.

SOLUTION

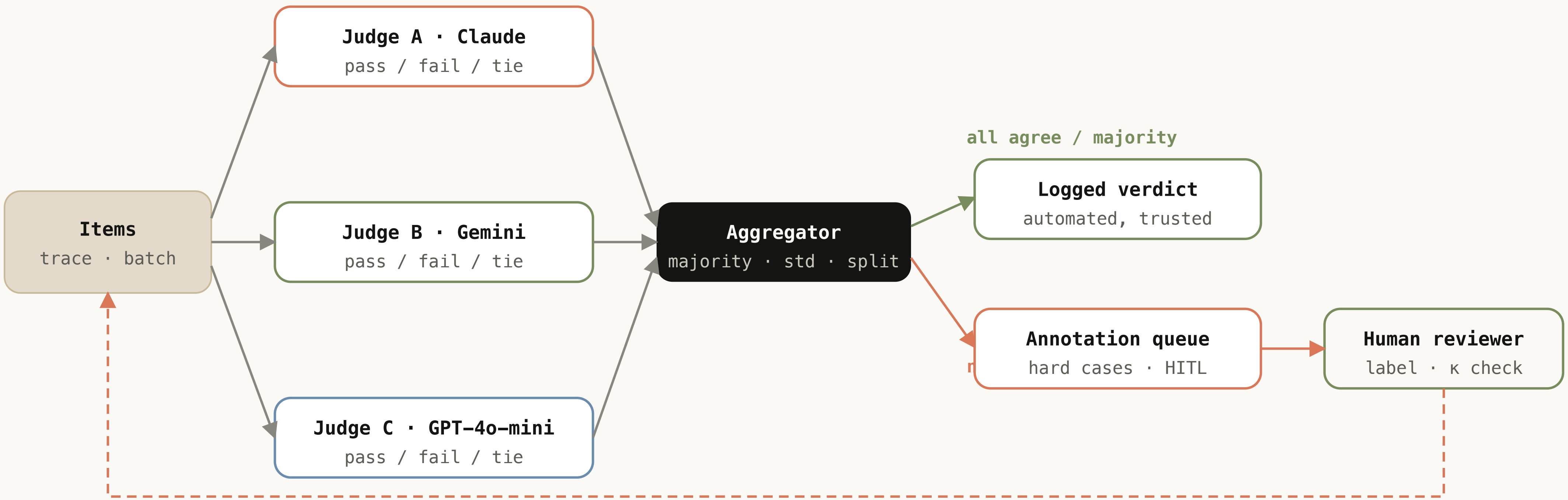
Panel of judges · run in parallel, aggregate.

Verga et al. 2024 · three small models from three providers (PoLL) beat one large judge on six datasets at ~7× lower cost.



Hard cases route to humans.

The panel's real output isn't the verdict — it's the routing decision. Easy cases auto-log; splits escalate. Human labels feed back into the rubric.



BASELINE

The ceiling is human–human, not 1.0.

Demanding perfection from judges flatters bad work. Humans don't agree perfectly either.

Target κ ≈ 0.65 — lower edge of the substantial-agreement band. Conservative. A panel that exceeds it is production-ready.

HOW TO CHOOSE

Selection is an optimisation problem.

Don't pick "the best models." Pick the cheapest subset that hits the agreement target with diverse error profiles.

```
# Formal framing
minimize:     cost(judges)
subject to:   κ(judges, human) ≥ 0.65
constraint:   no two judges from same provider family
```

Tornede et al. 2025 · multi-objective HPO. "Don't Pick the Highest-Performing Model" 2026 · greedy mutual-information selection.

DEFAULTS

The minimal-jury checklist.

DECISION	DEFAULT
Panel size	**3 (odd)** — avoids ties, plateau at 3–5
Provider mix	**One per family** — Anthropic + OpenAI + Google
Judge tier	**Small / fast** — panel of small can beat one large
Aggregation (boolean)	k-of-N majority
Aggregation (numeric)	Mean + report std
Aggregation (categorical)	Mode + explicit `TIE` label
Aggregation (pairwise)	Majority over A / B / TIE
Position swap	For pairwise only
Per-judge params	Don't expose · one prompt, temp 0

LESSONS FROM PRODUCTION

What keeps showing up.

SETUP

Make TIE a real label

Don't collapse 1:1:1 splits to null. TIE = ambiguous sample or unclear prompt — that's data.

SETUP

Position swap for pairwise

Re-run hard A-vs-B as B-vs-A and average. Cheap fix for position bias.

OPTIMISATION

Self-judge exclusion

If your generator is Claude, your judges cannot include Claude. Sibling-model agreement is fake.

OPTIMISATION

Provider > model diversity

Three different families > three different OpenAI models. Mix providers first.

THE ASK

Build the jury.

Start with the minimum viable version · three judges, one real prompt, pairwise mode. Don't overthink it. The first run surfaces position bias, provider quirks, TIE behaviour — failure modes metrics alone won't catch.

- **Run judges in parallel.** Concurrent API calls.
- **Cache repeated prompts.** Save cost during iteration.
- **Log disagreement explicitly.** Don't average too early.
- **Monitor per-judge κ** against a small labeled holdout.
- **Iterate on provider mix.** Not prompt engineering.

CLOSE

Weak judges. Diverse errors. Route the splits.

3 providers, ~7× cheaper, human-level agreement. The rest is operational.

↗ full blog ↗ interactive prototype ↗ workflow design orq.ai